# Project Documentation: Polynomial Regression on Manufacturing Dataset

## 1. Project Overview

This project applies **Polynomial Regression** (a form of linear regression where the relationship between the independent variable and dependent variable is modeled as an nth-degree polynomial) to analyze manufacturing metrics and predict the **Quality Rating** based on process variables like temperature and pressure.

## 2. Dataset Description

The dataset used in this project is manufacturing.csv, containing 3,957 records with the following features:

- **Temperature (°C)**: The operating temperature during manufacturing.
- **Pressure (kPa)**: The operating pressure during the process.
- **Temperature x Pressure**: An interaction term between temperature and pressure.
- **Material Fusion Metric**: A recorded metric associated with material fusion.
- **Material Transformation Metric**: A recorded metric associated with material transformation.
- **Quality Rating**: The target variable representing the final quality score of the manufactured product.

## 3. Step-by-Step Implementation Workflow

### Step 1: Environment Setup & Library Imports

Essential data science, machine learning, and visualization libraries are imported to handle preprocessing, model training, evaluation, and plotting.

- *Libraries used*: numpy, pandas, matplotlib.pyplot, seaborn, sklearn (train_test_split, PolynomialFeatures, StandardScaler, LinearRegression, mean_squared_error, r2_score).

### Step 2: Data Loading and Inspection

The dataset is loaded directly from the Kaggle input directory using pandas.read_csv(). Initial inspection verifies the first few rows, column data types, and checks for any missing values to ensure data integrity.

### Step 3: Exploratory Data Analysis (EDA)

A correlation matrix heatmap is generated using Seaborn to understand the linear relationships and multi-collinearity between the numerical features and the target variable (Quality Rating).

### Step 4: Feature and Target Selection

- **Features ($X$)**: Selected as Temperature (°C) and Pressure (kPa).

- **Target ($y$)**: Selected as Quality Rating.

### Step 5: Data Splitting

The dataset is partitioned into training and testing sets using an 80-20 split (test_size=0.2), ensuring proper evaluation on unseen data (X_train, X_test, y_train, y_test).

### Step 6: Polynomial Feature Transformation & Scaling

- **Polynomial Features**: PolynomialFeatures with degree=2 is applied to capture non-linear relationships and interaction terms between the features.
- **Standardization**: StandardScaler is utilized to normalize the transformed features, improving numerical stability and convergence during training.

### Step 7: Model Training

A standard LinearRegression model is fitted on the polynomial-transformed and scaled training data (X_train_scaled, y_train).

### Step 8: Model Evaluation

The model's predictive performance is measured on the test set (X_test_scaled) using standard regression metrics:

- **Root Mean Squared Error (RMSE)**: ~10.4180
- **Mean Squared Error (MSE)**: ~108.5342
- **R-squared ($R^2$) Score**: ~0.4684

### Step 9: Visualization

A scatter plot comparing **Actual vs. Predicted Quality Rating** alongside a reference identity line ($y = x$) is rendered to visually analyze model accuracy and error distribution.

## 4. Conclusion & Next Steps

The degree-2 polynomial model captures baseline non-linear trends in the manufacturing data. To further improve the $R^2$ score and reduce error, future iterations can experiment with:

1. Tuning the polynomial degree (degree=3 or higher).
2. Incorporating additional features like Material Fusion Metric or interaction columns.
3. Trying alternative regression algorithms (e.g., Ridge, Lasso, or tree-based regressors).